

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION
CLASS IX • SCIENCE (SUBJECT CODE: 086) • ACADEMIC SESSION 2026-27

SET F — RAPID FIRE SPEED TEST (30 MINS)

CHAPTER 05: EXPLORING MIXTURES: SOLUTION, COLLOID & SUSPENSION
TIME ALLOWED: 30 MINUTES MAXIMUM MARKS: 25 MARKS SERIES: CBS-IX-SCI-SET-F-05

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET F - RAPID FIRE SPEED TEST):

1. The question paper comprises **25 Rapid Fire Multiple Choice Questions** of **1 mark each**.
2. Designed for **Rapid Recall & High-Speed Accuracy**: Recommended target time is **30 Minutes**.
3. All questions are compulsory. There is no negative marking.
4. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

 **SPEED CHALLENGE INSTRUCTIONS:** Answer promptly without looking at notes. Test your instant recall of formulas, scientific definitions, and core NCERT concepts!

Q.1 What is the typical particle size in a true solution?

- (A) Less than 1 nm ($< 10^{-9}$ m) (B) 1 nm to 100 nm
(C) Greater than 100 nm (D) Greater than 1 mm

Q.2 Which of the following is a homogeneous mixture?

- (A) Brass alloy (B) Chalk powder in water
(C) Muddy pond water (D) Smoke in air

Q.3 The scattering of a visible light beam by colloidal particles is called the:

- (A) Doppler effect (B) Tyndall effect
(C) Raman effect (D) Photoelectric effect

Q.4 Milk is an example of which colloidal system?

- (A) Aerosol (B) Emulsion (liquid dispersed in liquid)
(C) Solid sol (D) Foam

Q.5 What is the continuous random zigzag motion of colloidal particles called?

- (A) Periodic motion (B) Brownian motion
(C) Translational drift (D) Circular motion

Q.6 Which separation method is based on the difference in boiling points of miscible liquids?

- (A) Filtration (B) Distillation
(C) Centrifugation (D) Chromatography

Q.7 To separate dyes present in black ink, which technique is utilized?

- (A) Paper chromatography (B) Sedimentation
(C) Magnetic separation (D) Sublimation

Q.8 Which of the following undergoes sublimation upon heating?

- | | |
|--|---------------------------------------|
| (A) Ammonium chloride (NH_4Cl) | (B) Sodium chloride (NaCl) |
| (C) Copper sulfate | (D) Calcium carbonate |
-

Q.9 What is tincture of iodine?

- | | |
|-------------------------------|---------------------------------|
| (A) Iodine dissolved in water | (B) Iodine dissolved in alcohol |
| (C) Iodine in milk | (D) Liquid iodine gas |
-

Q.10 Which of the following is a chemical change?

- | | |
|----------------------|------------------------------|
| (A) Melting of ice | (B) Rusting of iron |
| (C) Boiling of water | (D) Dissolving salt in water |
-

Q.11 What is the mass concentration of a solution containing 40 g of salt in 360 g of water?

- | | |
|---------|-----------|
| (A) 10% | (B) 11.1% |
| (C) 40% | (D) 9% |
-

Q.12 Which separation technique is used in dairies to separate cream from milk?

- | | |
|--------------------|--------------------|
| (A) Centrifugation | (B) Filtration |
| (C) Evaporation | (D) Chromatography |
-

Q.13 Fog and cloud are examples of:

- | | |
|-----------------------------|---------------|
| (A) Aerosol (liquid in gas) | (B) Solid sol |
| (C) Foam | (D) Gel |
-

Q.14 A solution in which no more solute can dissolve at that particular temperature is called:

- | | |
|--------------------------|------------------------|
| (A) Unsaturated solution | (B) Saturated solution |
| (C) Superheated solution | (D) Colloidal solution |
-

Q.15 What happens to the solubility of most solids (like sugar or KNO_3) in water as temperature increases?

- | | |
|-----------------------|------------------|
| (A) Decreases | (B) Increases |
| (C) Remains unchanged | (D) Becomes zero |
-

Q.16 What happens to the solubility of gases (like oxygen) in liquids as temperature increases?

- | | |
|----------------------|---------------|
| (A) Increases | (B) Decreases |
| (C) Remains constant | (D) Doubles |
-

Q.17 Which method is best suited to obtain pure copper sulfate crystals from an impure sample?

- | | |
|-----------------------|----------------------------|
| (A) Crystallization | (B) Evaporation to dryness |
| (C) Simple filtration | (D) Centrifugation |
-

Q.18 Two immiscible liquids having different densities can be separated using a:

- | | |
|--------------------------|----------------------|
| (A) Separating funnel | (B) Liebig condenser |
| (C) Fractionating column | (D) Centrifuge tube |
-

Q.19 Which of the following is classified as a pure chemical substance?

- | | |
|---|---------------|
| (A) Pure distilled water (H_2O) | (B) Air |
| (C) Brass | (D) Petroleum |
-

Q.20 Shaving cream is a colloid belonging to the category of:

- (A) Foam (gas in liquid) (B) Aerosol
(C) Solid sol (D) Emulsion

Q.21 Jelly and cheese are examples of which colloidal state?

- (A) Gel (liquid dispersed in solid) (B) Sol
(C) Foam (D) Emulsion

Q.22 Which process is used to separate the different elemental gases present in liquid air?

- (A) Fractional distillation (B) Chromatography
(C) Magnetic separation (D) Crystallization

Q.23 Burning of a candle involves:

- (A) Only a physical change (B) Only a chemical change
(C) Both physical and chemical changes (D) Neither physical nor chemical change

Q.24 Which component of air is present in the highest percentage by volume?

- (A) Oxygen (21%) (B) Nitrogen (78%)
(C) Argon (0.9%) (D) Carbon dioxide (0.04%)

Q.25 When sulfur dissolves in carbon disulfide (CS_2), which is the solute and which is the solvent?

- (A) Sulfur is solute, CS_2 is solvent (B) CS_2 is solute, Sulfur is solvent
(C) Both are solutes (D) Both are solvents

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET F)

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & RAPID EXPLANATIONS (SET F)

Q. No.	Correct	Scientific Rationale & Formula Summary
Q.1	(A)	Particles in a true solution are individual ions or molecules smaller than 1 nanometer ($< 10^{-9}$ m).
Q.2	(A)	Brass is a solid homogeneous solution of zinc in copper.
Q.3	(B)	John Tyndall discovered that colloidal particles scatter light, illuminating the beam path (Tyndall effect).
Q.4	(B)	In milk, liquid fat globules are dispersed in an aqueous liquid medium, forming an emulsion.
Q.5	(B)	Brownian motion is the random zigzag movement of colloidal particles resulting from continuous collisions with solvent molecules.
Q.6	(B)	Simple and fractional distillation separate miscible liquids that have distinct boiling points.
Q.7	(A)	Paper chromatography separates soluble pigments based on differential solubility and adsorption rates.
Q.8	(A)	Ammonium chloride, camphor, naphthalene, and iodine sublime directly from solid to vapor upon heating.
Q.9	(B)	Tincture of iodine is an antiseptic solution of 2–7% elemental iodine in ethyl alcohol and water.
Q.10	(B)	Rusting of iron forms a chemically new compound (hydrated ferric oxide), which is irreversible.
Q.11	(A)	$\text{Mass \%} = \frac{40}{40 + 360} \times 100 = \frac{40}{400} \times 100 = 10\%$.
Q.12	(A)	Centrifugation spins the milk rapidly, forcing denser skim milk to the outside while lighter fat cream stays at the center.
Q.13	(A)	In fog and clouds, liquid water droplets are dispersed in a gaseous dispersion medium (air).
Q.14	(B)	A saturated solution holds the maximum possible equilibrium amount of dissolved solute at a given temperature.
Q.15	(B)	Dissolution of most solid salts is endothermic; increasing temperature increases solubility.
Q.16	(B)	Dissolution of gases in liquids is exothermic; heating expels dissolved gas molecules, decreasing solubility.
Q.17	(A)	Crystallization purifies heat-sensitive salts without thermal decomposition or trapping soluble contaminants.
Q.18	(A)	A separating funnel allows the denser lower liquid to be drained via a stopcock from the lighter floating liquid.
Q.19	(A)	Pure distilled water is a chemical compound with a fixed definite proportion of hydrogen and oxygen (1:8 by mass).
Q.20	(A)	Shaving cream consists of air gas bubbles dispersed within a soapy liquid medium (foam).
Q.21	(A)	In a gel, a liquid is trapped and dispersed inside a continuous solid network.
Q.22	(A)	Fractional distillation of liquefied air separates nitrogen (-196°C), argon (-186°C), and oxygen (-183°C).
Q.23	(C)	Melting and vaporizing wax are physical changes; combustion of wax vapor into CO_2 and H_2O is a chemical change.
Q.24	(B)	Nitrogen makes up approximately 78.08% of dry atmospheric air by volume.
Q.25	(A)	Solid sulfur dissolves in liquid non-polar carbon disulfide; sulfur is solute and CS_2 is solvent.